Assignment Autumn 2026

COMP1013 Analytics Programming

Due Monday 1 JUNE 2026 (Week 14)

1 Project Description

In this assignment there are 4 parts. For each part you should:

- Explain what the task is asking to demonstrate your understanding of the requirement.
- Explain your rationales behind choices made answering the tasks.
- Ensure data wrangling is done correctly.
- Write the appropriate R code correctly.
- Include comments within the code to explain the algorithm.
- Test the code to ensure its correctness.
- Format and structure the code to maximise its readability.
- Provide appropriate interpretation of the results.

A report must be submitted containing a cover page, the solutions to each of the four parts, and your code, as a **PDF**, to the vUWS submission site. You must also follow a good practice on how to use GIT and make sure you do regular commit and push the code to the remote git repo regularly.

The cover page must contain your name, student number, subject code and name, and the declaration below.

Submissions in other formats or without cover pages will have marks deducted.

Late submissions will receive a 10% reduction in marks for each day late.

2 Marking Criteria

This assignment is worth 40% of the subject assessment tasks. There are four problems to investigate and 10 marks available for each of the four problems. In addition, there is 10 marks for using GIT in the assignment. Therefore, the **total marks** for assignment is **50**. The marking criteria for each question is given in Table 2.

Criteria	Q1	Q2	Q3	Q4
Explanation of task & your assumption (1 mark)				
Data wrangling & Code correctness (4 marks)				
Comments explaining code (1 marks)				
Code Testing (1 mark)				
Code Style and Readability (1 mark)				
Interpretation of results/analysis (2 marks)				
Total (10 marks)				

Table 1: Marking criteria for each part of this project.

For GIT, the following marking criteria will be used:

Criteria	GIT
Basic GIT command (3 marks) An appropriate number of Commit and Commit messages (3 marks) Advanced GIT (4 marks)	
Total (10 marks)	

Table 2: Marking criteria for GIT part of this project.

When writing the solutions to each of the four questions, make sure to consult the marking criteria and check that you have covered them. The project will be marked using this criteria.

For each task, there are a maximum of 2 bonus marks available for answers above and beyond the subject content. For example, extra analysis.

3 Declaration

Before submitting the assignment, include the following declaration in a clearly visible and readable place on the cover page of your project report.

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By including this statement, we the authors of this work, verify that:

- We hold a copy of this assignment that we can produce if the original is lost or damaged.
- We hereby certify that no part of this assignment/product has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.
- No part of this assignment/product has been written/produced for us by another person except where such collaboration has been authorised by the subject lecturer/tutor concerned.
- We are aware that this work may be reproduced and submitted to plagiarism detection software programs for the purpose of detecting possible plagiarism (**which may retain a copy on its database for future plagiarism checking**).
- We hereby certify that we have read and understand what the School of Computing, Engineering and Mathematics defines as minor and substantial breaches of misconduct as outlined in the learning guide for this unit.

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Note: An examiner or lecturer/tutor has the right not to mark this project report if the above declaration has not been added to the cover of the report.

Note: No AI (regarless of type, including agentic AI) use is allowed. You can use extra library you think it is useful in your assignment. Please also be reassured that the data used in this assignment is syntethic records.

4 Project Tasks

You are working at Play Pulse Studios as a data scientist and analyst. You are tasked to analyse the online game data based on demography. The data set is contained in three different sets:

players: Player demographic and account information:

- `player_id` – The unique identifier of the player.
- `age` – The age of the player.
- `country` – The country where the player is located.
- `signup_date` – The date the player registered their account.

games: Game metadata and classification:

- `game_id` – The unique identifier of the game.
- `game_name` – The name of the game.
- `genre` – The genre category of the game (e.g., Action, RPG, Strategy).

- `platform` – The platform on which the game is played (e.g., PC, Console, Mobile).

sessions: Player gameplay session records:

- `session_id` – The unique identifier of each gameplay session.
- `player_id` – The identifier linking the session to a player.
- `game_id` – The identifier linking the session to a game.
- `play_time_minutes` – The duration of the session in minutes.
- `score` – The score achieved by the player during the session.

Your tasks are:

1. Player Engagement Analysis by Experience Level

   Write the code to analyse player behaviour across different experience levels. The players should be grouped into 3 groups based on their `signup_date`:

   - Veteran: before 2023
   - Intermediate: 2023
   - New: after 2023

   For each group, calculate:

   - Number of players
   - Average play time (minutes)
   - Average score

   Tabulate the data using `kable` or `kableExtra`.

   Visualise the Average Play Time by Player Group using `ggplot2`.

   You are required to ensure that NA values are handled appropriately in your analysis.

   Explain your findings.

2. Game Genre Analysis

   Write the code to analyse player engagement across different game genres.

   Calculate for each genre:

   - Average play time
   - Average score
   - Number of sessions
   - Number of unique players

   Tabulate the results using `kable` or `kableExtra`.

   Visualise the Number of Unique Players by Genre using `ggplot2`.

   You are required to ensure that NA values are handled appropriately in your analysis.

   Discuss your findings. Which genre appears to be the most popular and engaging?

3. Top Players and Their Behaviour

   Write the code to analyse the top players and their gaming behaviour.

   First, identify the top 10 players based on total play time.

   For these top 10 players, calculate:

   - Total number of sessions
   - Average play time
   - Average score

   Tabulate the summary using `kable` or `kableExtra`.

   Visualise the score distribution of these players using a boxplot in `ggplot2`.

   You are required to ensure that NA values are handled appropriately in your analysis.

   Discuss your findings.

4. Age Group Behaviour Comparison

   Write the code to analyse whether there is a difference in gaming behaviour across age groups.

   Create the following age groups:

   - 16–25
   - 26–35
   - 36–45
   - 46+

   For each age group, calculate:

   - Average play time
   - Average score
   - Number of sessions

   Visualise the Average Play Time by Age Group using `ggplot2`.

   You are required to ensure that NA values are handled appropriately in your analysis.

   Discuss your findings. Are younger players more active or higher scoring?